Inverse-Root Support and Fourier Fusion for Polynomial Exponential Periods

Unconditional cyclicity, complete decomposition laws, and an exact-arithmetic atlas through degree twenty

Anonymous

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Status: unrefereed theorem candidate. The universal support and fusion claims rest on the displayed proofs. The degree-at-most-twenty atlas is a finite computer-assisted theorem candidate, supported by 1,152 exact-arithmetic case records under the stated SymPy trust boundary. No unaffiliated specialist review, independent external reproduction, proof-assistant formalisation, or settled novelty and priority determination has occurred.

Abstract

Let $P \in \boldsymbol{C}[x]$ have degree $d \geq 2$, and let

$$\mathcal{H}_P = \frac{\boldsymbol{C}(s)[x]\,dx}{\{(\partial_x f + sP'f)\,dx : f \in \boldsymbol{C}(s)[x]\}}$$

with connection

$$\nabla_s[f\,dx] = [(\partial_s f + Pf)\,dx].$$

Write $q(P, dx)$ for the dimension of the cyclic subspace generated by the fixed class $[dx]$. After affine normalisation, let

$$P(\xi_P(t)) = t^d, \qquad \xi_P(t) = t + O(t^{-1})$$

be the unique inverse-at-infinity Laurent series. Split it into congruence channels

$$\xi_P(t) = \sum_{r \in \boldsymbol{Z}/d\boldsymbol{Z}} X_r(t), \qquad X_r(t) = \sum_{k \equiv r \pmod d} a_k t^k,$$

and put $\Sigma(P) = \{r : X_r \neq 0\}$. We construct an explicit formal moment transform at $s = 0$ and prove that it is an isomorphism of differential modules

$$\mathcal{H}_P \otimes_{\boldsymbol{C}(s)} \boldsymbol{C}((s)) \simeq \bigoplus_{r=1}^{d-1} \mathcal{K}_r,$$

where $\mathcal{K}_r$ is the rank-one Kummer channel of exponent $-r/d$. The fixed seed has a nonzero component in $\mathcal{K}_r$ exactly when $r \in \Sigma(P)$. A leading-term Wronskian then gives the unconditional identity

$$q(P, dx) = |\Sigma(P)| = \dim_{\boldsymbol{C}} W_P^{\mathrm{red}},$$

where W_P^{red} is the reduced constant span of the inverse roots of $P(X) - z$. This removes stationary phase from the fixed-seed rank theorem. We also construct fixed rapid-decay contours on every sufficiently narrow s-sector and compute their Fourier–Gamma period-matrix asymptotic. The resulting matrix is non-singular, so differentiation, integration by parts, and separation of de Rham classes hold without a moving-cycle assumption or a general duality theorem.

The same support gives a complete block-fusion law. Suppose $P = R \circ Q$, with $m = \deg Q$, $n = \deg R$, and $d = mn$. For $c \in \boldsymbol{Z}/m\boldsymbol{Z}$, define

$$L_c = \{\ell \in \boldsymbol{Z}/n\boldsymbol{Z} : c + m\ell \in \Sigma(P)\}.$$

If V_a is the constant span of the roots in the a-th Q-block, then, for every set $A \subseteq \boldsymbol{Z}/n\boldsymbol{Z}$,

$$\dim \sum_{a \in A} V_a = \sum_{c\colon L_c \neq \varnothing} \operatorname{rank}\left(\zeta_n^{a\ell}\right)_{a \in A,\ \ell \in L_c}.$$

This formula determines every block-sum dimension, equality, pairwise intersection, partial fusion, and full fusion. Equality classes of block spans form a monodromy-invariant coarsening and therefore define a canonical actual polynomial right factor. Along a complete decomposition, projected support ranks form a monotone chain; at global rank at most three, no more than two levels can create rank. Every Ritt move replaces one intermediate projection by another and obeys an exact fibre-product constraint. At rank at most three, exactly eleven unordered Ritt rank diamonds are possible. For

$$P = (x^4 + x)^3 = H(x^3), \qquad H(y) = y(y+1)^3,$$

one has $\Sigma(P) = \{1, 7, 10\} \subset \boldsymbol{Z}/12\boldsymbol{Z}$. Projection modulo four gives the profile $[1, 1, 1]$, so the three quartic block spans coincide. Projection modulo three gives $[3]$, so the four cubic block lines span the same three-space. This is the complete linear explanation of the Ritt collision.

The theorem classifies block fusion for every polynomial and every actual polynomial decomposition. It does not, by itself, enumerate all affine families that realise a prescribed support set; that inverse realisability problem is separate.

We also give a finite affine atlas for (q(P,dx)le3) in degrees at most twenty. A fail-closed exact-arithmetic Gröbner sweep records every one of the 1,152 support cases in degrees five through twenty; degrees two through four are resolved directly. The all-degree atlas remains conjectural. Its first uncovered quartic lift has degree thirty-two, and the supplied general descent argument contains an identified circular step. Finally, we separate proved consequences from applications: generic period-ODE order is a corollary, whereas the zero-cycle and Laurent-polynomial formulations are research directions rather than closed theorems.

Keywords: polynomial exponential periods; twisted de Rham module; inverse roots; polynomial decomposition; Ritt moves; discrete Fourier transform; computer-assisted classification.

中文摘要

令 P 為次數 $d \geq 2$ 的複係數多項式，並以正規化的無窮遠反函數 $P(\xi(t)) = t^d$ 定義非零同餘通道集合 $\Sigma(P)$。本文建立明確的形式 矩變換，將固定種子 $[dx]$ 分解為秩一的 Kummer 通道，從而無條件證明 $q(P, dx) = |\Sigma(P)| = \dim_{\boldsymbol{C}} W_P^{\mathrm{red}}$。對每一個實際多項式 分解 $P = R \circ Q$，所有根區塊和的維數由有限 Fourier 子矩陣的秩精確給出； 由此得到區塊相等、兩兩交集、融合、典範右因子、完整分解鏈及十一種低秩 Ritt 菱

形。另以失敗即停止的精確算術 Gröbner 掃描，逐案記錄五至二十次 的 1,152 個低秩支撐情形，並直接處理二至四次。全次數係數圖譜仍為猜想； 首個尚未覆蓋的四次提升出現在三十二次，既有下降論證的循環步驟已明確 保留。一般週期微分方程階數是帶泛型條件的推論，而零循環與 Laurent 多項式部分僅作為後續研究方向。

關鍵詞： 多項式指數週期；扭曲 de Rham 模；反根；多項式分解；Ritt 變換；離散 Fourier 變換；電腦輔助分類。

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1. Claim boundary and relation to the parent releases

The parent release *Fixed-Seed Cyclicity Loci for Polynomial Exponential Periods* proves the finite twisted-reduction calculus, exact fixed-seed Krylov strata, and several quartic and quintic certificates. It states the passage from quotient relations to contour periods only for horizontal cycles on which differentiation and integration by parts are justified [EP1].

The child release *Fixed-seed cyclic rank as reduced inverse-root span* proves, conditional on an imported stationary-phase bridge, that

$$q(P, dx) = \dim W_P^{\mathrm{red}},$$

classifies the indecomposable phases with $q \leq 3$, proves right-factor heredity, and identifies equivariant fusion under Ritt moves as the remaining decomposable frontier [EP2].

The lineage is additive, not retrospective. The following table states the claim bridge used by this candidate.

Earlier immutable output	Material inherited here	Change in this candidate
Fixed-Seed Cyclicity Loci for Polynomial Exponential Periods [EP1]	finite twisted reduction and exact Krylov computations	the analytic qualification is retained historically but is no longer needed for the algebraic rank theorem
Fixed-seed cyclic rank as reduced inverse-root span [EP2]	inverse-root span, right-factor heredity, and the low-rank frontier	the imported stationary-phase bridge is replaced by Theorem 4.3, and the former fusion obstruction is resolved by Theorem 8.2
Unconditional fixed-seed cyclic rank and Ritt-invariant block fusion, v0.2 source history	the complete proofs and eleven-diamond table	promoted into the flagship text, with notation and assurance boundaries made explicit
Affine coefficient families for inverse-root rank at most three, v0.3 source history	candidate normal forms and exact controls	retained only as a theorem through degree twenty; the all-degree assertion is withdrawn to Conjecture 13.2

This note makes five changes.

1. It replaces the imported stationary-phase bridge by a direct formal moment isomorphism. The rank equality is therefore algebraic and unconditional.
2. It constructs a fixed sectorial rapid-decay basis and proves an explicit non-singular Fourier–Gamma asymptotic for its period matrix. This closes the contour, differentiation, and integration-by-parts conditions directly in one variable.
3. It replaces the case-by-case fusion frontier by one Fourier-matrix formula that applies to every right factor, every complete decomposition, and every Ritt move.
4. It proves that equality fusion exposes a canonical polynomial factor, bounds rank-creating levels in low-rank decomposition chains, and gives the complete eleven-row rank table for adjacent low-rank Ritt moves.
5. It adds a finite, per-case exact-arithmetic coefficient atlas through degree twenty while preserving the unresolved all-degree descent as a conjecture rather than importing the circular argument from the atlas draft.

The word **unconditional** refers to these mathematical implications. It does not mean externally refereed, formally verified, or priority-certified.

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2. Algebraic setup and affine normalisation

Let $K = \boldsymbol{C}(s)$. For a polynomial P of degree $d \geq 2$, define

$$D_P(f) = \partial_x f + sP' f$$

and

$$\mathcal{H}_P = K[x]dx/D_P(K[x])dx.$$

The leading coefficient of sP' is a unit of K. Repeated cancellation of the leading monomial gives a unique representative of degree below $d-1$. Hence

$$\mathcal{H}_P \cong K^{d-1}$$

with basis

$$[dx], [x\,dx], \ldots, [x^{d-2}dx]. \qquad (2.1)$$

The connection

$$\nabla_s[f\,dx] = [(\partial_s f + Pf)\,dx] \qquad (2.2)$$

is well defined because

$$[\partial_s + P, \partial_x + sP'] = 0.$$

For the constant seed, $\nabla_s^j[dx] = [P^j dx]$. We define

$$q(P, dx) = \dim_K K\langle\nabla_s\rangle[dx]. \qquad (2.3)$$

An invertible affine source change rescales dx by a nonzero constant. A target scaling is absorbed by a change of s, while a target translation replaces powers of P by an invertible triangular combination. Thus $q(P, dx)$ is invariant under invertible affine source and target changes.

We may therefore assume throughout that

$$P(x) = x^d + \sum_{j=0}^{d-2} p_j x^j. \qquad (2.4)$$

For convenience we also subtract $P(0)$. This target translation does not change any rank below. Equation (2.4) makes P monic and centred.

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3. The inverse-at-infinity support

Proposition 3.1 — Normalised inverse at infinity

There is a unique Laurent series

$$\xi(t) = t + \sum_{k \le 0} a_k t^k \in t\,\boldsymbol{C}[[t^{-1}]] \qquad (3.1)$$

such that

$$P(\xi(t)) = t^d. \qquad (3.2)$$

Because P is centred, $a_0 = 0$, so

$$\xi(t) = t + O(t^{-1}). \qquad (3.3)$$

Proof

Put $y = t^{-1}$ and $\xi = tF(y)$, where $F(0) = 1$. Equation (3.2) becomes

$$F^d + \sum_{j=0}^{d-2} p_j y^{d-j} F^j = 1. \qquad (3.4)$$

At each positive power y^N, the new coefficient of F enters through F^d with coefficient d, while every other term depends only on previous coefficients. Characteristic zero gives a unique recursive solution. The coefficient of t^{d-1} in $P(\xi)$ is da_0, hence $a_0 = 0$. ▫

Let $\zeta_d = e^{2\pi i/d}$. For $r \in \boldsymbol{Z}/d\boldsymbol{Z}$, put

$$X_r(t) = \frac{1}{d} \sum_{j=0}^{d-1} \zeta_d^{-jr}\, \xi(\zeta_d^j t) = \sum_{k \equiv r \,(\mathrm{mod}\, d)} a_k t^k. \qquad (3.5)$$

Define the **inverse support**

$$\Sigma(P) = \{r \in \boldsymbol{Z}/d\boldsymbol{Z} : X_r \neq 0\}. \qquad (3.6)$$

The branches of $P(X) - z$ near infinity, with $z = t^d$, are

$$x_j(z) = \xi(\zeta_d^j t) = \sum_{r \in \boldsymbol{Z}/d\boldsymbol{Z}} \zeta_d^{jr} X_r(t). \qquad (3.7)$$

The average of the branches is zero because P is centred. Thus

$$X_0 = 0, \qquad 0 \notin \Sigma(P). \qquad (3.8)$$

Proposition 3.2 — Affine invariance of reduced support data

After removal of the trace channel, invertible affine source and target changes preserve $\Sigma(P)$ up to multiplication of its indices by a unit in $\boldsymbol{Z}/d\boldsymbol{Z}$. In particular, they preserve its cardinality, every divisor-fibre partition, and all Fourier ranks used below.

Proof

A source translation adds the same constant to every inverse branch and hence changes only the trace channel; centring removes it. A source scaling multiplies every non-trace channel by a nonzero constant. A target scaling rescales t. For a target translation by c, the old and new infinity parameters satisfy

$$\tilde{t} = t(1 - ct^{-d})^{1/d} = t\,H(t^{-d}), \qquad H(0) = 1.$$

Substitution $t \mapsto tH(t^{-d})$ preserves each exponent residue modulo d, and its inverse has the same form, so no nonzero channel can appear or disappear. Finally, changing the generator of the cyclic inertia group multiplies the residue labels by a unit. ▫

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4. A direct formal moment transform

The purpose of this section is to prove the rank theorem without contour integrals and without stationary phase.

Put $K_0 = \boldsymbol{C}((s))$. For $r \in \{1, \dots, d-1\}$, let

$$\mathcal{K}_r = K_0 e_r$$

be the rank-one differential module with

$$\partial_s e_r = -\frac{r}{d} s^{-1} e_r. \qquad (4.1)$$

For every $n \in \boldsymbol{Z}$, define nonzero constants $\mu_{r,n}$ by

$$\mu_{r,0} = 1, \qquad \mu_{r,n+1} = -\frac{r + nd}{d}\mu_{r,n}. \qquad (4.2)$$

No factor in (4.2) vanishes because $1 \le r \le d-1$.

Consider a Laurent one-form, bounded above, whose exponents lie in one congruence class:

$$\omega = \sum_{n \le N} b_n t^{r+nd-1} dt.$$

Define

$$\mathfrak{M}_r(\omega) = \left(\sum_{n \le N} b_n \mu_{r,n} s^{-n} \right) e_r \in \mathcal{K}_r. \qquad (4.3)$$

This is a well-defined Laurent series in s: only finitely many negative powers occur. The pullbacks used below have coefficients with a uniform lower bound on s-adic valuation. Thus (4.3) extends continuously and $\boldsymbol{C}((s))$-linearly to those pullbacks; no uncontrolled double Laurent series is introduced.

Lemma 4.1 — Formal integration by parts

For every bounded-above Laurent series $h(t)$,

$$\mathfrak{M}_r\Big(\big(\partial_t h + s d t^{d-1} h\big)\,dt\Big) = 0 \qquad (4.4)$$

in each congruence channel.

Proof

It is enough to take $h = t^{r+nd}$. The two terms in (4.4) map to

$$(r+nd)\mu_{r,n} s^{-n} e_r$$

and

$$d\mu_{r,n+1} s^{-n} e_r.$$

Their sum is zero by (4.2). Termwise application is legitimate for a series bounded above. □

Pull a polynomial form back by $x = \xi(t)$:

$$\rho(f(x)dx) = f(\xi(t))\xi'(t)dt. \qquad (4.5)$$

Because $P(\xi) = t^d$,

$$\rho\big((\partial_x q + sP'q)dx\big) = \big(\partial_t(q(\xi)) + s d t^{d-1} q(\xi)\big)dt. \qquad (4.6)$$

Lemma 4.1 therefore makes the direct sum of the moment maps descend to

$$\mathfrak{M}\colon \mathcal{H}_P \otimes_K K_0 \to \bigoplus_{r=1}^{d-1} \mathcal{K}_r. \qquad (4.7)$$

Lemma 4.2 — Compatibility with the s-connection

The map $\mathfrak{M}$ is a morphism of differential modules.

Proof

For a monomial channel,

$$\begin{aligned} \partial_s\big(\mu_{r,n} s^{-n} e_r\big) &= -\Big(n + \frac{r}{d}\Big)\mu_{r,n} s^{-n-1} e_r \\ &= \mu_{r,n+1} s^{-n-1} e_r. \end{aligned} \qquad (4.8)$$

The right-hand side is the moment of multiplication by t^d. Under $P(\xi) = t^d$, this is exactly the action of P, while coefficient derivatives obey the ordinary product rule. Hence

$$\mathfrak{M} \circ \nabla_s = \partial_s \circ \mathfrak{M}.$$

□

Theorem 4.3 — Formal moment isomorphism

The map (4.7) is an isomorphism.

Proof

Use the basis (2.1). From (3.3), for $0 \le j \le d-2$,

$$\xi(t)^j \xi'(t) = t^j + O(t^{j-2}). \qquad (4.9)$$

Write the matrix of $\mathfrak{M}$ after identifying the target basis with $e_1, \dots, e_{d-1}$. In row r, the coefficient of $s^0 e_r$ is the coefficient of $t^{r-1}dt$ in (4.9). It vanishes when $r > j+1$, and equals one when $r = j+1$. Terms with lower powers of t contribute positive powers of s; no negative power can occur because $j+1 < d$.

Thus the matrix lies in $\mathrm{Mat}_{d-1}(\boldsymbol{C}[[s]])$, and its constant term is triangular with every diagonal entry equal to one. Its determinant is a unit. Both sides of (4.7) have rank $d-1$ over K_0, so $\mathfrak{M}$ is an isomorphism. □

This theorem is the missing analytic bridge in algebraic form. It is stronger for the present purpose than a stationary-phase rank comparison: it gives an explicit channel-by-channel differential-module isomorphism and requires no cycle, convergence, Stokes, or contour hypothesis.

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5. The unconditional fixed-seed rank theorem

Write

$$\xi(t) = \sum_{k \le 1} a_k t^k.$$

The pullback of the fixed seed is

$$\xi'(t)dt = \sum_{k \le 1} k a_k t^{k-1} dt. \qquad (5.1)$$

For $r \in \{1, \dots, d-1\}$, its r-th moment component is

$$v_r = \sum_{n \in \boldsymbol{Z}} (r+nd) a_{r+nd} \mu_{r,n} s^{-n} e_r. \qquad (5.2)$$

The sum is bounded below as a Laurent series in s. Since $r + nd \neq 0$ and $\mu_{r,n} \neq 0$, distinct coefficients in (5.2) occupy distinct powers of s. Therefore

$$v_r \neq 0 \quad \Leftrightarrow \quad X_r \neq 0 \quad \Leftrightarrow \quad r \in \Sigma(P). \qquad (5.3)$$

Lemma 5.1 — Active Kummer channels are cyclically independent

Let $S \subseteq \{1, \dots, d-1\}$, and let $v = (v_r)_{r \in S}$ with every $v_r \neq 0$. Then the cyclic differential submodule generated by v in $\bigoplus_{r \in S} \mathcal{K}_r$ has rank $|S|$.

Proof

For each r, let n_r be the largest index with nonzero coefficient in (5.2). Then

$$v_r = c_r s^{-n_r} e_r(1 + O(s)), \qquad c_r \neq 0. \qquad (5.4)$$

Put

$$\alpha_r = -n_r - \frac{r}{d} = -\frac{r + n_r d}{d}.$$

The numbers α_r are distinct because their numerators are distinct modulo d. For $j \geq 0$,

$$\partial_s^j v_r = c_r (\alpha_r)_{\underline{j}} s^{-n_r - j} e_r(1 + O(s)), \qquad (5.5)$$

where $(u)_{\underline{j}} = u(u-1)\cdots(u-j+1)$. The determinant of the first $|S|$ derivative columns has leading coefficient

$$\left(\prod_{r \in S} c_r \right) \prod_{r < r'} (\alpha_{r'} - \alpha_r), \qquad (5.6)$$

up to a nonzero monomial in s. This is nonzero. Thus the first $|S|$ derivatives are independent. The ambient active direct sum has rank $|S|$, so the cyclic rank is exactly $|S|$. □

Theorem 5.2 — Unconditional support formula

For every polynomial $P \in \boldsymbol{C}[x]$ of degree at least two,

$$q(P, dx) = |\Sigma(P)|. \qquad (5.7)$$

Proof

Affine-normalise P. By Theorem 4.3, extending scalars from K to K_0 identifies the cyclic submodule generated by $[dx]$ with the cyclic submodule generated by the vector (v_r). Scalar extension preserves rank. Equation (5.3) identifies the active summands with $\Sigma(P)$, and Lemma 5.1 gives their rank. □

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6. Equality with the reduced inverse-root span

Inside the local splitting field $\boldsymbol{C}((t^{-1}))$, with $z = t^d$, let W_P^{red} be the constant $\boldsymbol{C}$-span of the inverse roots of $P(X) - z$, with the trace line removed. Passing to the corresponding algebraic-function field does not change this constant dimension. In centred coordinates the trace is already zero, so we write W_P.

Theorem 6.1 — Root span as inverse support

$$\dim_{\boldsymbol{C}} W_P = |\Sigma(P)|. \qquad (6.1)$$

Proof

Equation (3.7) expresses the vector of roots $(x_0, \ldots, x_{d-1})$ as the discrete Fourier transform of $(X_0, \ldots, X_{d-1})$. The Fourier matrix is invertible. The nonzero X_r are linearly independent over $\boldsymbol{C}$, because their Laurent supports occupy disjoint residue classes modulo d. Hence the root span has dimension equal to the number of nonzero channels. Equation (3.8) removes the trace channel. □

Combining Theorems 5.2 and 6.1 gives the main unconditional identity

$$q(P, dx) = |\Sigma(P)| = \dim_{\mathbf{C}} W_P^{\mathrm{red}}. \qquad (6.2)$$

The conductor equality in the child release remains a valid independent representation-theoretic explanation. It is no longer needed to establish (6.2).

Corollary 6.2 — Inverse support is the infinity-character spectrum

Let τ be the full cycle at infinity, indexed by $\tau(x_j) = x_{j+1}$. Then

$$\tau(X_r) = \zeta_d^r X_r. \qquad (6.3)$$

Hence $\Sigma(P)$ is exactly the set of nontrivial inertia characters that occur in W_P^{red}, each with multiplicity one. Replacing the generator of inertia multiplies the indices by a unit modulo d. The projection laws for polynomial right factors are therefore character restriction laws for subgroups of the cyclic inertia group.

Proof

Apply τ to the Fourier inverse $X_r = d^{-1} \sum_j \zeta_d^{-jr} x_j$ and reindex. Distinct channels are one-dimensional by their disjoint Laurent supports. □

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7. Explicit analytic realisation without a moving-cycle gap

The algebraic rank theorem is complete without period integrals. This section constructs the required rapid-decay cycles explicitly and proves that their period matrix is non-singular.

Assume that

$$P(x) = x^d + O(x^{d-2})$$

is monic and centred. Fix $\theta_0 \in \boldsymbol{R}$ and $0 < \varepsilon < \pi/2$, and let

$$\mathfrak{S} = \{s \in \boldsymbol{C}^* : |\arg s - \theta_0| < \varepsilon\}. \qquad (7.1)$$

For $k \in \boldsymbol{Z}/d\boldsymbol{Z}$, put

$$\phi_k = \frac{\pi - \theta_0 + 2\pi k}{d}. \qquad (7.2)$$

Let ρ_k be the ray $x = re^{i\phi_k}$, $r \geq 0$, and let Γ_k run from infinity to the origin along ρ_k, then from the origin to infinity along ρ_{k+1}.

Proposition 7.1 — Uniform rapid decay

For every compact subsector $K \Subset \mathfrak{S}$, there are constants $R, c, C > 0$ such that

$$\Re(sP(re^{i\phi_k})) \leq -cr^d + C \qquad (7.3)$$

for every $s \in K$, every $r \geq R$, and every k.

Proof

By construction,

$$\Re(se^{id\phi_k}) = -|s|\cos(\arg s - \theta_0).$$

On a compact subsector this is bounded above by a negative constant. The centred lower terms are $O(r^{d-2})$, uniformly in the finitely many chosen directions, and are absorbed by half of the leading negative term for large r. □

For every polynomial A, the integral

$$I_{k,A}(s) = \int_{\Gamma_k} A(x)e^{sP(x)}\,dx \qquad (7.4)$$

therefore converges absolutely and locally uniformly. The chains are fixed throughout $\mathfrak{S}$, so differentiation produces no chain-velocity term.

Proposition 7.2 — Differentiation and twisted exactness

For every $j \geq 0$,

$$\partial_s^j I_{k,A}(s) = \int_{\Gamma_k} P(x)^j A(x)e^{sP(x)}\,dx. \qquad (7.5)$$

For every polynomial f,

$$\int_{\Gamma_k} (\partial_x f + sP'f)e^{sP(x)}\,dx = 0. \qquad (7.6)$$

Proof

On compact subsectors, Proposition 7.1 dominates every differentiated integrand by $C(1+r)^N e^{-cr^d}$. Dominated differentiation proves (7.5). For (7.6), the integrand is $d(fe^{sP})$. The two finite endpoints at the origin cancel, and rapid decay kills both ends at infinity. □

Radially compactify the affine line. The d chosen decay directions are marked boundary points. The resulting relative-chain model gives

$$H_1^{\mathrm{rd}}(\boldsymbol{A}^1, e^{sP}) \simeq \tilde{H}_0(\{\rho_0, \ldots, \rho_{d-1}\}; \boldsymbol{C}), \qquad (7.7)$$

with $[\Gamma_k] \mapsto e_{k+1} - e_k$. Hence $[\Gamma_0], \ldots, [\Gamma_{d-2}]$ form a basis and $\sum_{k=0}^{d-1} [\Gamma_k] = 0$.

For $0 \leq k, j \leq d-2$, define

$$\Pi_{k,j}(s) = \int_{\Gamma_k} x^j e^{sP(x)}\,dx. \qquad (7.8)$$

Theorem 7.3 — Fourier–Gamma period determinant

Let $\zeta_d = e^{2\pi i/d}$ and put $r = j + 1$. As $s = \rho e^{i\vartheta_0} \to 0$ on the central ray,

$$\lim_{\rho \downarrow 0} s^{r/d} \Pi_{k,j}(s) = \frac{\Gamma(r/d)}{d} e^{\pi i r/d} (\zeta_d^r - 1) \zeta_d^{kr}. \qquad (7.9)$$

Consequently

$$\det \Pi(s) = C_d s^{-(d-1)/2} (1 + o(1)), \qquad C_d \neq 0. \qquad (7.10)$$

Proof

Choose the branch of $s^{1/d}$ on $\mathfrak{S}$ and put $u = s^{1/d} x$. Then

$$sP(x) = u^d + \sum_{j=0}^{d-2} p_j s^{1-j/d} u^j.$$

On each scaled ray, the leading term has real part $-|u|^d$. Young's inequality absorbs every lower term into a fixed fraction of $|u|^d$, so dominated convergence applies. The scaled ray angles are $\psi_k = (\pi + 2\pi k)/d$. The integral from the origin to infinity on a ray of angle ψ is

$$\frac{1}{d} \Gamma(r/d) e^{ir\psi}.$$

Taking the oriented difference of the two rays gives (7.9). Its constant matrix is a non-singular diagonal matrix multiplied by

$$(\zeta_d^{kr})_{0 \le k \le d-2,\ 1 \le r \le d-1},$$

which is a Vandermonde matrix on the distinct numbers $\zeta_d, \dots, \zeta_d^{d-1}$. Finally, $\sum_{r=1}^{d-1} r/d = (d-1)/2$, proving (7.10). □

Corollary 7.4 — Unconditional period interpretation

The period matrix $\Pi(s)$ is invertible throughout $\mathfrak{S}$. Every relation in $\mathcal{H}_P$ therefore gives the corresponding differential relation for the complete rapid-decay period system, and no additional relation vanishes against all of these cycles.

Proof

Theorem 7.3 gives invertibility at sufficiently small points of the central ray. The sector $\mathfrak{S}$ is simply connected and excludes $s = 0$. In the polynomial basis (2.1), Proposition 7.2 gives $\partial_s \Pi = \Pi A(s)$ for the connection matrix $A(s) \in \mathrm{Mat}_{d-1}(\boldsymbol{C}(s))$. Invertibility at one point therefore makes Π a fundamental solution matrix throughout the sector. Equivalently, Jacobi's formula gives

$$\partial_s \det \Pi = \mathrm{tr}(A(s)) \det \Pi.$$

The unique solution with a nonzero initial value is an exponential of a holomorphic primitive times that value, and cannot vanish inside $\mathfrak{S}$. □

This directly closes convergence, differentiation, integration by parts, and separation of classes. General rapid-decay duality and stationary phase are consistent external frameworks, but neither is load-bearing here.

Corollary 7.5 — Universal monodromy at $s = 0$

Analytic continuation once anticlockwise around $s = 0$ sends

$$[\Gamma_k] \mapsto [\Gamma_{k-1}] \qquad (7.11)$$

with indices modulo d. In the basis $[\Gamma_0], \ldots, [\Gamma_{d-2}]$, one has

$$[\Gamma_{d-1}] = -\sum_{j=0}^{d-2} [\Gamma_j].$$

The monodromy characteristic polynomial is therefore

$$\frac{\lambda^d - 1}{\lambda - 1}, \qquad (7.12)$$

and its eigenvalues are ζ_d^{-r}, $1 \le r \le d-1$. These are exactly the monodromies of the Kummer channels in Theorem 4.3. This computes the sectorial continuation relevant at $s = 0$. Stokes matrices at $s = \infty$ contain additional critical-value data, but are not hypotheses for the rank or period-relation theorems.

A chosen single cycle can still annihilate a nonzero differential submodule of the fixed-seed cyclic module. Its scalar period has minimal order $q(P, dx)$ over $\boldsymbol{C}(s)$ exactly when the induced differential-module map on that cyclic module has zero kernel, equivalently when no nonzero class in the module has identically zero period on the chosen cycle. This is a non-annihilation condition on that cycle, not an analytic hypothesis.

---

8. Block spans for a polynomial decomposition

Suppose

$$P = R \circ Q, \qquad m = \deg Q, \qquad n = \deg R, \qquad d = mn. \qquad (8.1)$$

After affine changes between the factors, assume that Q, R, and P are monic and that Q is centred. Then P is centred as well.

Index the roots of $P(X) - t^d$ by

$$x_j = \xi(\zeta_d^j t), \qquad j \in \boldsymbol{Z}/d\boldsymbol{Z}. \qquad (8.2)$$

Since

$$Q(x_j) \sim x_j^m \sim \zeta_n^j t^m,$$

the Q-blocks are

$$B_a = \{a + nb : b \in \boldsymbol{Z}/m\boldsymbol{Z}\}, \qquad a \in \boldsymbol{Z}/n\boldsymbol{Z}. \qquad (8.3)$$

Let

$$V_a = \operatorname{span}_{\boldsymbol{C}}\{x_j : j \in B_a\} \subseteq W_P. \tag{8.4}$$

For $c \in \boldsymbol{Z}/m\boldsymbol{Z}$, define

$$L_c = \{\ell \in \boldsymbol{Z}/n\boldsymbol{Z} : c + m\ell \in \Sigma(P)\} \tag{8.5}$$

and

$$E_c = \operatorname{span}_{\boldsymbol{C}}\{X_{c+m\ell} : \ell \in L_c\}. \tag{8.6}$$

The spaces E_c form a direct sum because they have disjoint Laurent supports modulo m.

Theorem 8.1 — Fourier block decomposition

For every $a \in \boldsymbol{Z}/n\boldsymbol{Z}$,

$$V_a = \bigoplus_{c : L_c \neq \varnothing} \boldsymbol{C}\, Y_{a,c}, \tag{8.7}$$

where

$$Y_{a,c} = \sum_{\ell \in L_c} \zeta_n^{a\ell} X_{c+m\ell} \in E_c. \tag{8.8}$$

A harmless nonzero factor ζ_d^{ac} has been omitted from (8.8).

Proof

Using (3.7), for $b \in \boldsymbol{Z}/m\boldsymbol{Z}$,

$$\begin{aligned} x_{a+nb} &= \textstyle\sum_{r \in \Sigma(P)} \zeta_d^{(a+nb)r} X_r \\ &= \textstyle\sum_c \zeta_m^{bc} \zeta_d^{ac} \sum_{\ell \in L_c} \zeta_n^{a\ell} X_{c+m\ell}. \end{aligned} \tag{8.9}$$

The discrete Fourier transform in b is invertible and isolates one value of c at a time. Thus the span of the block roots is the span of the vectors in (8.8). Vectors for different c lie in the direct summands E_c. Each $Y_{a,c}$ is nonzero because the $X_{c+m\ell}$ are independent. This proves (8.7). □

Theorem 8.2 — Complete block-fusion rank formula

For every subset $A \subseteq \boldsymbol{Z}/n\boldsymbol{Z}$,

$$\dim \sum_{a \in A} V_a = \sum_{c : L_c \neq \varnothing} \operatorname{rank}\left(\zeta_n^{a\ell}\right)_{a \in A,\, \ell \in L_c}. \tag{8.10}$$

Proof

Inside E_c, the vectors $Y_{a,c}$ have coefficient rows $(\zeta_n^{a\ell})_{\ell \in L_c}$ in the independent basis $(X_{c+m\ell})_{\ell \in L_c}$. Their span therefore has the displayed matrix rank. The summands E_c are direct, so their dimensions add. □

Equation (8.10) determines every partial-sum dimension, coincidence, and rank increase among the blocks. Pairwise intersections follow by the dimension identity in Corollary 8.4. We make no separate claim here that every higher intersection lattice is literally the rank of one Fourier submatrix.

Corollary 8.3 — Exact right-factor heredity

Every block has dimension

$$\dim V_a = \#\{c : L_c \neq \varnothing\}. \qquad (8.11)$$

Since one block is the root span of $Q(X) - y$ after a transcendental base change,

$$q(Q, dx) = \dim W_Q = |\pi_m(\Sigma(P))|, \qquad (8.12)$$

where $\pi_m : \boldsymbol{Z}/d\boldsymbol{Z} \to \boldsymbol{Z}/m\boldsymbol{Z}$ is reduction modulo m. In particular, $q(Q, dx) \leq q(P, dx)$.

Corollary 8.4 — Pairwise intersections

For $a, b \in \boldsymbol{Z}/n\boldsymbol{Z}$,

$$\dim(V_a \cap V_b) = \#\left\{c : L_c \neq \varnothing,\ (a-b)(\ell - \ell') \equiv 0 \pmod{n}\ \forall \ell, \ell' \in L_c\right\}. \qquad (8.13)$$

Indeed, in each E_c, the two one-dimensional lines coincide exactly when their coefficient vectors are proportional.

Corollary 8.5 — Equality stabiliser

Define

$$H_m(P) = \left\{\delta \in \boldsymbol{Z}/n\boldsymbol{Z} : \delta(\ell - \ell') \equiv 0 \pmod{n} \quad \forall c,\ \ell, \ell' \in L_c\right\}. \qquad (8.14)$$

Then $H_m(P)$ is a subgroup and

$$V_a = V_b \quad \Leftrightarrow \quad b - a \in H_m(P). \qquad (8.15)$$

The number of distinct block spans is

$$[\boldsymbol{Z}/n\boldsymbol{Z} : H_m(P)]. \qquad (8.16)$$

Corollary 8.6 — Full fusion

The following are equivalent:

1. all block spans coincide;
2. every block span equals W_P;
3. $q(P, dx) = q(Q, dx)$;
4. every nonempty L_c has size one;
5. the projection $\pi_m : \Sigma(P) \to \boldsymbol{Z}/m\boldsymbol{Z}$ is injective.

Thus full fusion has an exact character-theoretic criterion.

Corollary 8.7 — No fusion

The block spaces $V_0, \dots, V_{n-1}$ form a direct sum if and only if every nonempty L_c equals all of $\boldsymbol{Z}/n\boldsymbol{Z}$. Equivalently,

$$q(P, dx) = nq(Q, dx). \qquad (8.17)$$

Every intermediate fusion pattern is measured by the Fourier ranks in (8.10).

---

9. All complete decompositions and Ritt moves

Let

$$P = P_1 \circ P_2 \circ \cdots \circ P_k \qquad (9.1)$$

be a complete decomposition into indecomposable polynomials. For each suffix

$$Q_i = P_i \circ \cdots \circ P_k, \qquad m_i = \deg Q_i, \qquad (9.2)$$

the corresponding block system and every fusion datum are obtained by projecting the one fixed support set

$$\Sigma(P) \subseteq \boldsymbol{Z}/d\boldsymbol{Z}$$

modulo m_i and applying Theorem 8.2.

This already handles every partial decomposition: any right factor can be refined to a complete suffix. A full cycle of length d has a unique block system of each specified block size, so two right factors of the same degree induce the same partition at infinity, up to relabelling.

Ritt's first theorem states that any two complete decompositions of a complex polynomial are connected by adjacent Ritt moves and inserted affine changes. Affine changes preserve the reduced support data. In a nontrivial adjacent move, two coprime indecomposable degrees a and b are interchanged. If the fixed suffix to their right has degree h, the move changes the one intermediate suffix degree

$$bh \leftrightarrow ah, \qquad (9.3)$$

while keeping h and abh fixed.

Theorem 9.1 — Ritt-invariant fusion classification

The complete block-fusion data of P, across every complete decomposition and every sequence of Ritt moves, are the collection of Fourier rank systems

$$\mathcal{F}_m(P) = \left\{ \operatorname{rank}\left(\zeta_{d/m}^{a\ell}\right)_{a \in A,\, \ell \in L_c} : A \subseteq \boldsymbol{Z}/(d/m)\boldsymbol{Z},\ c \in \boldsymbol{Z}/m\boldsymbol{Z} \right\} \qquad (9.4)$$

for the suffix degrees m occurring in the Ritt decomposition graph. Every $\mathcal{F}_m(P)$ is computed from the same invariant $\Sigma(P)$. Under an adjacent move (9.3), the fusion system $\mathcal{F}_{bh}(P)$ is replaced by $\mathcal{F}_{ah}(P)$; all other levels remain unchanged.

Proof

Theorem 8.2 gives $\mathcal{F}_m(P)$ for each actual suffix degree. Ritt's first theorem connects all complete decompositions by moves that change only one intermediate suffix degree. Since the total polynomial and its inverse support do not change, the stated projection replacement is exact. Conversely, every block equality or rank in any decomposition is one of the ranks in (9.4). □

When $\gcd(a,b)=1$, the Chinese remainder theorem views the characters in a moved ab-block on a two-coordinate grid. The monomial and Chebyshev forms in Ritt's second theorem determine when the polynomial move exists; once it exists, Theorem 9.1 determines the accompanying fusion by reading the same characters along the other coordinate projection. The linear classification therefore requires no separate monomial, Chebyshev, or exceptional fusion case.

9.2 Canonical fusion quotient

Let $\mathcal{B}$ be the block system induced by a decomposition $P=R\circ Q$. For a block B, put

$$V(B)=\operatorname{span}_{\boldsymbol{C}}\{x_i : i\in B\},$$

and declare $B\sim B'$ when $V(B)=V(B')$.

Theorem 9.2 — Equality fusion exposes an actual factor

The unions of the $\sim$-classes form a monodromy-invariant block system. Consequently there is a polynomial factorisation

$$P=R^{\sharp}\circ Q^{\sharp},\qquad Q^{\sharp}=S\circ Q,\qquad (9.5)$$

unique up to an affine change of intermediate variable, whose blocks are exactly those unions. Distinct $Q^{\sharp}$-blocks have distinct root spans. If $m=\deg Q$, $n=\deg R$, and $H_m(P)$ is (8.14), then

$$\deg Q^{\sharp}=m/|H_m(P)|,\qquad \deg R^{\sharp}=n/|H_m(P)|.\qquad (9.6)$$

Proof

Let G be the geometric monodromy group of $P(X)-z$. For every $g\in G$, linearity gives $gV(B)=V(gB)$. Thus $B\sim B'$ if and only if $gB\sim gB'$. The equivalence classes of Q-blocks are permuted by G, so their unions form a G-invariant block system on the roots of $P(X)-z$.

By the block-system/intermediate-field correspondence, this coarsening gives a field

$$\boldsymbol{C}(P)\subseteq L\subseteq \boldsymbol{C}(Q)\subseteq \boldsymbol{C}(x).$$

Lüroth's theorem writes $L=\boldsymbol{C}(Q^{\sharp})$ for a rational function $Q^{\sharp}$. Polynomial monodromy is totally ramified over infinity: the inertia generator is a full d-cycle, and the induced inertia action on every intermediate block system is again a full cycle. Consequently the intermediate cover has one point above infinity. After a Möbius change of its target, $Q^{\sharp}$ has its only pole at $x=\infty$, hence is a polynomial. The inclusion $L\subseteq\boldsymbol{C}(Q)$ then yields $Q^{\sharp}=S\circ Q$; the same one-pole argument permits S to be chosen polynomial. Finally $P\in L$, so $P=R^{\sharp}\circ Q^{\sharp}$ with $R^{\sharp}$ polynomial.

Two Lüroth generators for L differ by a Möbius transformation. Since both polynomial generators have their unique pole at infinity, that transformation fixes infinity and is affine. This proves the asserted uniqueness. In the cyclic infinity indexing, Corollary 8.5 identifies the equality classes with the cosets of $H_m(P)$. Each union contains $m|H_m(P)|$ roots, which gives (9.6). □

Repeated equality of block spans is thus never merely a numerical accident. It identifies a canonical hidden factor, although full fusion can produce only the trivial factor $Q^\sharp = P$.

For a partial-fusion control, view

$$P(x) = x^8 + x^4$$

through the right factor $Q = x^2$. Its support is $\{1, 5\}$ modulo eight. The four Q-block lines occur in two equal pairs, with $H_2(P) = \{0, 2\} \subset \boldsymbol{Z}/4\boldsymbol{Z}$. The theorem recovers the coarser actual right factor $Q^\sharp = x^4$, of degree $2|H_2(P)| = 4$, and an outer quadratic.

9.3 Fusion defects along a decomposition chain

For a complete decomposition (9.1), let Q_i and m_i be as in (9.2), and put

$$r_i = q(Q_i, dx) = |\pi_{m_i}\Sigma(P)|. \qquad (9.7)$$

Then $r_1 \geq r_2 \geq \cdots \geq r_k$. Define $\delta_i = r_i - r_{i+1}$. Projection gives

$$q(P, dx) - q(P_k, dx) = \sum_{i=1}^{k-1} \delta_i. \qquad (9.8)$$

Moreover, $\delta_i = 0$ exactly when the level $Q_i = P_i \circ Q_{i+1}$ has full fusion.

Corollary 9.3 — Low-rank chain sparsity

If $q(P, dx) \leq 3$, every complete decomposition has at most two rank-increasing levels above a rightmost power, at most one above a rightmost Chebyshev factor, and none above a rightmost rank-three quartic. Every other level is a rank-preserving full-fusion extension.

9.4 Ritt diamonds

Consider an adjacent nontrivial Ritt move that interchanges coprime degrees a, b above a fixed suffix of degree h. Put

$$r_0 = |\pi_h\Sigma(P)|, \quad r_a = |\pi_{ah}\Sigma(P)|, \quad r_b = |\pi_{bh}\Sigma(P)|, \quad r_2 = |\pi_{abh}\Sigma(P)|. \qquad (9.9)$$

Theorem 9.4 — Ritt fibre-product constraint

Let $S = \pi_{abh}\Sigma(P)$. For each occupied residue c modulo h, put

$$S_c = \{u \in S : u \equiv c \pmod h\}, \quad A_c = \pi_{ah}(S_c), \quad B_c = \pi_{bh}(S_c). \qquad (9.10)$$

The map $u \mapsto (u \bmod ah, u \bmod bh)$ is injective on each S_c. Hence

$$|S_c| \leq |A_c|\,|B_c| \quad \text{and} \quad r_2 \leq \sum_c |A_c|\,|B_c|. \qquad (9.11)$$

Proof

Equal images modulo ah and bh imply equality modulo $\operatorname{lcm}(ah, bh) = abh$, since $\gcd(a, b) = 1$. □

No universal additive inequality follows: for $(a, b, h) = (2, 3, 1)$ and $S = \{0, 1, 3, 4\}$, one has $(r_0, r_a, r_b, r_2) = (1, 2, 2, 4)$.

Theorem 9.5 — Complete low-rank Ritt table

If $r_2 \le 3$, the unordered pair $\{r_a, r_b\}$ is exactly one of the following eleven possibilities.

Top r_2	Bottom r_0	Intermediate ranks $\{r_a, r_b\}$
1	1	$\{1, 1\}$
2	1	$\{1, 2\}$
2	1	$\{2, 2\}$
2	2	$\{2, 2\}$
3	1	$\{1, 3\}$
3	1	$\{2, 2\}$
3	1	$\{2, 3\}$
3	1	$\{3, 3\}$
3	2	$\{2, 3\}$
3	2	$\{3, 3\}$
3	3	$\{3, 3\}$

Proof

The maps to residues modulo ah and bh jointly separate the elements of S. For one element the result is immediate. For two elements, two rank-one intermediate partitions cannot jointly separate them. For three elements, a rank-one intermediate forces the other to have rank three; over a two-block bottom partition, two rank-two refinements coincide and therefore cannot have discrete common refinement; and over a one-block bottom, two distinct rank-two partitions can have discrete common refinement. These cases are exactly the displayed rows. □

The table exhausts the rank behaviour of actual low-rank Ritt moves. It does not assert that each row is realised by a polynomial identity. For the collision in Section 11, $(a, b, h) = (3, 4, 1)$ and

$$(r_0; r_a \leftrightarrow r_b; r_2) = (1; 3 \leftrightarrow 1; 3). \qquad (9.12)$$

---

10. Complete low-rank fusion profiles

Suppose $q(P, dx) = |\Sigma(P)| \le 3$. For a fixed right-factor degree m, the nonzero fibre sizes $(|L_c|)$ form a partition of $q(P, dx)$. Hence the following list is exhaustive.

Global rank	Fibre-size profile	Right-factor rank $q(Q, dx)$	Block geometry
1	$[1]$	1	Every block line is the same line.
2	$[1,1]$	2	Every block plane equals W_P.
2	$[2]$	1	Block lines form a Fourier orbit spanning a plane.
3	$[1,1,1]$	3	Every block three-space equals W_P.
3	$[2,1]$	2	Block planes share the singleton channel; the other line moves in a two-channel plane.
3	$[3]$	1	Block lines form a Fourier orbit spanning a three-space.

The equality stabiliser (8.14) tells how many geometrically distinct blocks occur inside each profile. Thus the table is a complete structural classification, not merely a list of possible dimensions.

This table combines with the child release's indecomposable theorem as follows. Every indecomposable suffix of a rank-at-most-three phase is a power, a Chebyshev polynomial, or an indecomposable quartic. The table then describes all ways in which that suffix can sit inside every larger decomposition. What remains outside the present theorem is an affine **realisability atlas**: which coefficient families realise each admissible support and decomposition lattice.

---

11. The generating Ritt collision

Put

$$Q(x) = x^4 + x, \qquad P(x) = Q(x)^3. \qquad (11.1)$$

There are two decompositions:

$$P = y^3 \circ Q \qquad (11.2)$$

and

$$P = H \circ x^3, \qquad H(y) = y(y+1)^3. \qquad (11.3)$$

The inverse at infinity satisfies

$$\xi(t)^4 + \xi(t) = t^4, \qquad (11.4)$$

with expansion

$$\xi(t) = t - \frac{1}{4}t^{-2} - \frac{1}{32}t^{-5} + \cdots. \qquad (11.5)$$

Let $\omega^3 = 1$. Equation (11.4) gives

$$\xi(\omega t) = \omega \xi(t), \qquad (11.6)$$

so every support exponent is congruent to one modulo three. The quartic trace is zero, so no support exponent is congruent to zero modulo four. The deck symmetry first leaves the four classes

$$\{1, 4, 7, 10\} \pmod{12}.$$

Exact heredity modulo four gives $\pi_4 \Sigma(P) = \Sigma(x^4 + x) = \{1, 2, 3\}$. The class $4 \pmod{12}$ maps to $0 \pmod 4$, so it is excluded. The first three nonzero terms in (11.5) occupy the remaining residues

$$1, 10, 7 \pmod{12}.$$

No fourth permitted residue remains. Hence

$$\Sigma(P) = \{1, 7, 10\}. \qquad (11.7)$$

11.1 Quartic-block view

For the right factor Q, $m = 4$ and $n = 3$. Reduction modulo four sends

$$1 \mapsto 1, \qquad 7 \mapsto 3, \qquad 10 \mapsto 2.$$

Thus the fibre profile is

$$[1, 1, 1].$$

By Corollary 8.6, the three quartic block spans are all equal to W_P, and

$$q(P, dx) = q(Q, dx) = 3. \qquad (11.8)$$

11.2 Cubic-block view after the Ritt move

For the right factor x^3, $m = 3$ and $n = 4$. All three support residues map to one modulo three. Thus the profile is

$$[3].$$

Every cubic block has rank one, the four block lines are distinct, and their sum has rank three. Formula (8.13) gives zero intersection for distinct lines.

The Ritt move therefore changes

$$[1, 1, 1] \quad \leftrightarrow \quad [3] \qquad (11.9)$$

by replacing projection modulo four with projection modulo three. The global rank and support remain fixed. This is the precise mechanism that the naive inheritance bound missed.

12. Exact finite computation

The accompanying program `src/fusion_support.py` implements two independent sides of the main identity for trusted rational input.

1. It computes $q(P, dx)$ by exact twisted-Krylov reduction over $\boldsymbol{Q}(s)$, returning a nonzero independence minor and the first dependence relation when one occurs.
2. It recursively solves (3.4) for the inverse-at-infinity coefficients. It stops when it has witnessed $q(P, dx)$ distinct non-trace residue classes. Theorem 5.2 proves that no further class can occur.
3. For every divisor $m \mid d$ with $m \geq 2$, it computes the support fibres L_c, the right-factor rank predicted when a degree-m right factor exists, the equality stabiliser, pairwise intersection dimensions, and the full-fusion and direct-sum criteria.

The program does not assert that every divisor is an actual right-factor degree and does not enumerate the decomposition lattice. Standard polynomial decomposition algorithms can supply that lattice; the support formula then classifies every node and edge.

The frozen tests include:

Phase	Degree	Exact q	$\Sigma(P)$
$x^4 + x$	4	3	$\{1, 2, 3\}$
$(x^4 + x)^3$	12	3	$\{1, 7, 10\}$
$(x^3 + x)^2$	6	2	$\{1, 5\}$
$T_5 = 16x^5 - 20x^3 + 5x$	5	2	$\{1, 4\}$
$x^6 + x^2 + 1$	6	3	$\{1, 3, 5\}$
$(x^3 + x)^2 + (x^3 + x)$	6	4	$\{1, 2, 4, 5\}$

The degree-twelve receipt checks both Ritt views in Sections 11.1 and 11.2. The negative control raises the rank from two to four and changes the support fibres from $[1, 1]$ to $[2, 2]$.

A separate clean-room control in `cleanroom/fraction_audit.py` imports neither the production module nor a computer-algebra library. Exact `Fraction` arithmetic checks nine named phases, twelve deterministic random phases, four composition projections, the Ritt fibre-product constraint, and all eleven low-rank diamond types. It specialises the Krylov calculation at four nonzero rational values of s, so it is an independent fault control rather than a symbolic proof over $\boldsymbol{Q}(s)$.

---

13. Affine coefficient atlas through degree twenty

Write $P \sim \tilde{P}$ when the two phases differ by invertible affine changes in the source and target. Such changes preserve $q(P, dx)$ and the reduced inverse support up to multiplication of residue labels by a unit. We use the Dickson convention

$$D_d\left(u + \frac{a}{u}, a\right) = u^d + \left(\frac{a}{u}\right)^d. \qquad (13.1)$$

Theorem 13.1 — Finite low-rank atlas

Let $P \in \boldsymbol{C}[x]$ have degree $2 \le d \le 20$. Then $q(P, dx) \le 3$ if and only if P is affine-equivalent to a member of the following list. The rows are exact-rank rows after the stated lower-rank loci are removed.

Rank	Normal form	Parameter and degree conditions
1	x^d	$d \ge 2$
2	$D_d(x, a)$	$d \ge 3$, $a \ne 0$
2	$x^{2m} + Ax^m$	$m \ge 2$, $A \ne 0$
3	$x^{3m} + Ax^{2m} + Bx^m$	$m \ge 2$, excluding the loci below
3	$x^4 + Ax^2 + Bx$	$B \ne 0$
3	$(x^4 + ax)^3$	$a \ne 0$

In the cubic-over-power row, $A = B = 0$ has rank one. The only rank-two locus in that row is

$$m = 2, \qquad B = \frac{A^2}{4}, \qquad A \ne 0. \qquad (13.2)$$

For quartics, $B = 0$, $A \ne 0$ has rank two and $A = B = 0$ has rank one. The listed families cover the classification; they are not asserted to give unique parameter coordinates. Over $\boldsymbol{C}$, source scaling removes the nonzero parameter in the Dickson and quadratic-over-power rows. In the cubic-over-power and quartic rows the familiar scale-invariant ratios B/A^2 and A^3/B^2, where defined, describe the generic remaining moduli. The exceptional row is one affine orbit.

With the infinity generator normalised so that the leading inverse term has character 1, the realised supports are:

Coefficient family	$\Sigma(P)$
power, degree d	$\{1\}$
Dickson, degree d	$\{1, d-1\}$
quadratic over a power, degree $2m$	$\{1, m+1\}$
cubic over a power, degree $3m$	$\{1, m+1, 2m+1\}$
exact-rank quartic	$\{1, 2, 3\}$
exceptional degree-twelve collision	$\{1, 7, 10\}$

For $P = H(x^m)$, deck symmetry restricts the inverse exponents to $1 \bmod m$, and exact rank forces the displayed surviving classes. For Dickson polynomials, the parametrisation in (13.1) separates the characters 1 and -1. The quartic claim follows from the direct calculation below, and Section 11 proves the exceptional support. Thus the positive families are matched by complete support calculations, not only by truncated confinement.

Proof

Direct degrees two through four

After making P monic and centred and discarding a target constant, degree two is x^2. Degree three is $x^3 + px$: $p = 0$ is the power row, while $p \neq 0$ is $D_3(x, a) = x^3 - 3ax$. Every centred monic quartic is $x^4 + Ax^2 + Bx$. Exact reduction gives rank three exactly when $B \neq 0$, rank two when $B = 0$, $A \neq 0$, and rank one at $A = B = 0$. Thus no finite sweep is hidden behind the phrase “direct”.

Exact-arithmetic case construction

Put $y = t^{-1}$ and write

$$\xi(t) = tF(y), \qquad F(y) = 1 + \sum_{i \geq 2} u_i y^i.$$

The inverse equation is

$$F^d + \sum_{j=0}^{d-2} p_j y^{d-j} F^j = 1. \qquad (13.3)$$

For a proposed support $S \subseteq \{1, \ldots, d-1\}$, only the low coordinates u_i with $(1-i) \bmod d \in S$ are retained. Coefficients of (13.3) in orders $2, \ldots, d$ recover $p_{d-2}, \ldots, p_0$ triangularly. Higher forbidden coefficients give polynomial confinement equations. The implementation constructs the Gröbner basis from equations through order $4d$, checks every additional equation through order $6d$, and recomputes with the full set if stability fails.

More explicitly, once $p_{d-2}, \ldots, p_{d-n+1}$ are known, the coordinate map used by the engine is

$$p_{d-n} = -[y^n]\left(F^d + \sum_{j=d-n+1}^{d-2} p_j y^{d-j} F^j\right), \qquad 2 \leq n \leq d.$$

Substitution of each recorded component into this triangular map gives the coefficient normal forms in Theorem 13.1. This formula is also the exact source-to-encoding bridge implemented by the function build_constrained.

For each degree $d \geq 5$, the normalised class 1 must occur. The number of support sets of size at most three is therefore

$$1 + (d-2) + \binom{d-2}{2}. \qquad (13.4)$$

Summing (13.4) for $5 \leq d \leq 20$ gives 1,152 cases. Every case ends in one of three exact outcomes:

1. the confinement variety is the full allowed coordinate space and a saturating family has that dimension;
2. every non-power coordinate is nilpotent modulo the Gröbner ideal, with the reduction exponent recorded; or
3. a power of the product of equations defining the predicted components reduces to zero, proving set-theoretic containment in their union.

Every polynomial with support contained in S supplies inverse coordinates satisfying all confinement equations, hence lies in the truncated variety. The radical-containment witness places that variety inside the predicted component union, and the triangular map above sends those components to the listed coefficient families. Adding still more confinement equations can only shrink the variety, so emptiness and containment conclusions are truncation-safe. Conversely, the complete support calculations above show that each listed exact-rank family really occurs.

The frozen case inventory records the degree, support, truncation depths, predicted families and dimension, stability result, defining-ideal digest, Gröbner-basis digest, outcome type, reduction witness, Python version and SymPy version. It contains 45 full-family cases, 884 power-only cases and 223 union cases, with zero anomaly and zero missing certificate. Family members are checked against the computed ideals over a fixed 61-bit prime, while the load-bearing elimination and reductions are over $\boldsymbol{Q}$.

This is a finite computer-assisted theorem candidate, not a proof object in a proof assistant. Its trust boundary includes the inverse-series generator, the map from inverse coordinates to coefficient families, the case generator, Python, and SymPy's exact rational Gröbner arithmetic. The release runner executes the full sweep under ordinary and optimised Python, compares the two receipts byte for byte, and exits nonzero on either an anomaly or a missing certificate. Synthetic anomaly, missing-certificate, support, Fourier, receipt and manifest corruptions are required to fail. □

Conjecture 13.2 — All-degree atlas

The list in Theorem 13.1 is conjectured to be exhaustive in every degree. The candidate rank-preserving extensions are power chains, aligned Dickson chains, $(x^{3m}+bx^m)^2$, and $(x^4+ax)^3$.

The supplied all-degree draft does not prove this statement. Its quartic lift uses

$$A(x^m+b)-A(b)=Q_0(x)^m, \qquad (13.5)$$

but the directly derived fibre identity is a product of twisted quartic factors. The coefficient meant to force $m \mid 3$ cancels in that product unless the collapse to the single m-th power in (13.5) is already known. The analogous cubic step has the same defect. Exact elimination confirms the quartic conclusion for $2 \le m \le 7$; the first uncovered lift is $m=8$, of total degree thirty-two. A non-circular twisted-factor descent or a new monodromy argument is still required.

---

14. Consequences and research directions

14.1 Generic minimal ODE order: proved corollary

For a rapid-decay cycle Γ, put

$$I_\Gamma(s)=\int_\Gamma e^{sP(x)}\,dx.$$

Every differential relation in the cyclic module annihilates every such period. For a generic functional on the complete period solution space, Theorems 5.2 and 7.3 give minimal scalar differential-equation order

$$q(P, dx) = |\Sigma(P)|. \qquad (14.1)$$

The word “generic” is essential: a particular cycle can annihilate an active channel. The supplied exploratory reducer produces order-three operators for $(x^6 + x^2)^2$ and $(x^4 + x)^3$, while a generic degree-twelve phase has order eleven. Those demonstrations illustrate (14.1), but their numerical quadrature and fail-open script are not additional theorem evidence.

14.2 Zero-cycle Fourier channels: formulation, not closure

For weights n_j with zero sum, an identity

$$\sum_j n_j g(x_j(t)) = 0 \qquad (14.2)$$

is equivalent to vanishing of the corresponding Fourier channels of $g \circ \xi$. This formulation suggests exact tests for selected zero-cycle identities. In the degree-twelve collision, the supplied script identifies the candidate centre elements

$$g = x^7 + \frac{7}{4}x^4, \qquad g = x^{10} - \frac{5}{2}x^4.$$

No theorem in this package proves a complete zero-cycle criterion or resolves a named Gavrilov–Pakovich problem. The script prints `REVIEW` on a borderline comparison and returns success, so it is excluded from the release gate.

14.3 Laurent-polynomial extension: separate programme

For a chosen local branch of a Laurent polynomial, inverse-series channels turn residue moments into coefficient exclusions. The supplied small-bidegree experiments may be useful, but their control dimensions do not presently agree and the harness does not fail on the discrepancy. These materials are preserved under source history and are not evidence for a Laurent classification theorem.

---

15. Theorem-level relation to prior work

Sabbah [Sa99] places twisted de Rham complexes in a broad algebraic setting. Bloch and Esnault [BE04] and Hien [Hi09] establish rapid-decay/de Rham pairings for irregular connections; Hien and Roucairol [HR08] give integral representations for exponential Gauss–Manin systems. These works explain the analytic environment of Sections 2 and 7. The present claim is narrower and more explicit: for one-variable polynomial phases and the fixed class $[dx]$, Theorem 4.3 identifies the completed module channel by channel, and Theorem 5.2 reads its cyclic rank directly from inverse support. The targeted search found no statement of that fixed-seed formula in those sources.

Ritt [Ri22] supplies the decomposition moves used in Section 9, while Müller and Zieve [MZ08] give a modern structural treatment. Burda and Khovanskii [BK12] use primitive polynomial monodromy in a bounded-radical classification. The recent work of Behajaina, König and Neftin [BKN26] classifies a largeness alternative and explicit exceptions for monodromy groups of compositions of two indecomposable polynomials; it is directly relevant to a future all-degree proof of Conjecture 13.2. None of these sources, on the materials examined, states the all-block Fourier rank formula (8.10).

Girstmair [Gi99, Gi03] is a direct antecedent for constant relations among polynomial roots; the 2003 corrigendum is included. The Fourier support calculus concerns the reduced span of all local inverse roots and equality of entire decomposition blocks. It does not classify every possible root relation.

Álvarez Sánchez, Bravo Trinidad and Mardešić [ABM13] show, in the zero-dimensional Abelian-integral setting, that vanishing phenomena need not be sums of composition terms. Pakovich and Muzychuk [PM09] solve the related polynomial moment problem through decomposition data. These results motivate the caution in Section 14.2: equality fusion of whole block spans is not a complete zero-cycle theory.

The source audit used open full texts or arXiv versions for the main theorem comparisons and authoritative bibliographic pages to check metadata and corrigenda. It was targeted, not an exhaustive MathSciNet or zbMATH citation-network review. Non-location of an equivalent formula is therefore provisional evidence of originality, not proof of priority.

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16. Assurance, novelty, and remaining work

16.1 Closed in this candidate

Subject to ordinary mathematical checking of the displayed proofs, this note closes the following implications:

- the fixed-seed rank/root-span equality has no analytic or stationary-phase hypothesis;
- fixed sectorial rapid-decay contours give an unconditional analytic realisation, with an explicit non-singular Fourier–Gamma period matrix;
- block fusion for every actual decomposition is exactly the Fourier rank law (8.10);
- every block equality and pairwise intersection has the explicit criteria (8.13)–(8.16);
- all complete decompositions and Ritt moves are classified by congruence projections of one invariant support set;
- equality fusion exposes a canonical actual polynomial factor;
- rank-at-most-three chains have at most two rank-creation levels and every adjacent move lies in the eleven-row Ritt table; and
- the two faces of $(x^4 + x)^3$ are the profiles $[1, 1, 1]$ and $[3]$; and
- subject to the explicit computer-algebra trust boundary, the coefficient families in Theorem 13.1 exhaust $q(P, dx) \leq 3$ through degree twenty.

16.2 Not closed by this candidate

The following are distinct questions and are not claimed here:

- an all-degree affine coefficient-family theorem for $|\Sigma(P)| \leq 3$;
- a classification of which abstract support subsets are polynomially realisable;
- a new proof of Ritt's decomposition theorems;
- external specialist verification, formal verification, or independent reproduction;
- settled novelty, priority, or field-wide significance; and

- a guarantee that an arbitrarily chosen non-rapid-decay contour defines a period, or that every single rapid-decay contour is dual-cyclic.

The first two items form the remaining inverse-realisability problem. They are no longer needed to classify fusion once P and a decomposition are given, but they are required to extend Theorem 13.1 beyond degree twenty.

---

17. Reproducibility

The authoritative release command is

```
python3 run_core_checks.py
```

It first verifies complete manifest coverage. It then runs the structural tests, the independent-within-programme fraction audit, the period determinant spot check, exact-rank mutations, cross-implementation batteries, the quartic lift and extension-locus eliminations, the family-blind sparse search, the 1,152-case atlas sweep, and six semantic negative controls. The suite is repeated with `python3 -O`; generated ordinary and optimised receipts must be byte-identical and must match their frozen expected hashes.

To inspect only the generating example, run

```
python src/fusion_support.py --phase '(x**4 + x)**3'
```

During development, the bounded command

```
python3 run_core_checks.py --quick --development
```

does not replay the full atlas and is not a release gate. The `receipts/`, `structural-receipts/`, `cleanroom/`, and `verification/` directories retain the frozen outputs and the current-byte claim bridge. `ENVIRONMENT.txt` is the authoritative environment record. Exploratory ODE, zero-cycle and Laurent programs are excluded from the release command.

Data and code availability

The candidate package includes the manuscript sources and renderings, exact arithmetic programs, all case records, mutation controls, receipts, reviews, source history and a complete SHA-256 manifest. No external dataset is used.

Ethics declaration

This work concerns abstract mathematics and uses no human participants, animals, personal data or sensitive data.

Author contributions (CRediT)

Anonymous: Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing (original draft), Writing (review and editing). AI systems contributed to derivation, implementation, adversarial checking, literature routing and editorial synthesis. Those contributions do not constitute external peer review.

Conflict of interest

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AI disclosure

Generative AI systems were used extensively for mathematical exploration, code generation, proof drafting, adversarial checking, literature routing and editorial revision. The supplied full review was converted into a traceable revision and independently checked against the revised files. Model agreement, producer replay and separate code paths are not represented as peer review or unaffiliated reproduction.

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